CONTINUATION / EXERCISE 14/11

YOU CAN COMPUTE THE SAMPLES OF THE IMPULSE RESPONSE ALSO USING THE RECURSIVE EQUATION $\boxed{2^*}$

$$\boxed{y(t) = 0{,}3\, y(t-1) + 0{,}4\, y(t-2) + u(t-1) + 0{,}5\, u(t-2)}$$

$$u(t) = \begin{cases} 1 & t = 0 \\ 0 & t > 0 \end{cases}$$

IMPULSE RESPONSE = OUTPUT $y(t)$ WHEN THE SYSTEM IS FED BY $u(t)$

$$w(0) = y(0) = 0{,}3 \cdot y(-1) + 0{,}4 \cdot y(-2) + u(-1) + 0{,}5\, u(-2)$$

$$= 0 + 0 + 0 + 0 = \boxed{0}$$

ASSUMING NULL PAST VALUES OF $y(t)$

$$w(1) = y(1) = 0{,}3 \cdot y(0) + 0{,}4 \cdot y(-1) + u(0) + 0{,}5 \cdot u(-1)$$

$$= 0 + 0 + 1 + 0 = \boxed{1}$$

$$w(2) = y(2) = 0{,}3 \cdot y(1) + 0{,}4 \cdot y(0) + u(1) + 0{,}5 \cdot u(0)$$

$$= 0{,}3 + 0{,}5 = \boxed{0{,}8}$$

$$w(3) = y(3) = 0{,}3 \cdot y(2) + 0{,}4 \cdot y(1) + u(2) + 0{,}5 \cdot u(1)$$

$$= 0{,}24 + 0{,}4 = \boxed{0{,}64}$$

EXERCISE

CONSIDER THE FOLLOWING SAMPLES OF THE IMPULSE RESPONSE OF A LINEAR DYNAMIC SYSTEM:

$$w(0) = 0,\ w(1) = 0,\ w(2) = 1,\ w(3) = -1/2,\ w(4) = 1/4,\ w(5) = -1/8$$

$$w(6) = 1/16$$

⇒ COMPUTE THE SYSTEM TRANSFER FUNCTION

$$y(t) = \sum_{\tau=0}^{+\infty} w(\tau) \cdot u(t-\tau) = W(z) \cdot u(t)$$

LET'S EXPAND THE SERIES:

$$y(t) = w(0) \cdot u(t) + w(1) \cdot u(t-1) + w(2) \cdot u(t-2) + \dots$$

RECALL THE MEANING OF z^{-k}, BACKWARD SHIFT OPERATOR

$$= \left[w(0) + w(1) \cdot z^{-1} + w(2) \cdot z^{-2} + \dots \right] \cdot u(t)$$

$$= \left[1 \cdot z^{-2} - 1/2 \cdot z^{-3} + 1/4 \cdot z^{-4} + \dots \right] \cdot u(t)$$

$$= z^{-2} \left[\underbrace{1 - 1/2\, z^{-1} + 1/4\, z^{-2} + \dots}_{\sum_{\ell=0}^{+\infty} \left(-1/2 \cdot z^{-1}\right)^{\ell}} \right] \cdot u(t)$$

→ GEOMETRIC SERIES

$$= z^{-2} \cdot \frac{1}{1 - \left(-1/2\, z^{-1}\right)} \cdot u(t)$$

$$= \boxed{\frac{z^{-2}}{1 + 1/2\, z^{-1}}} \cdot u(t)$$

TRANSFER FUNCTION (FILTER FORM)

$$\underbrace{\frac{1}{z^2 + 1/2\, z}}$$

TRANSFER FUNCTION

⇛ COMPUTE A STATE-SPACE REPRESENTATION OF THE SYSTEM 4SID

STEP # 1 - ORDER OF THE SYSTEM

THE SYSTEM ORDER IS THE SIZE OF THE BIGGEST HANKEL MATRIX WHICH HAS THE FOLLOWING PROPERTIES:

- IT IS NOT NULL
- $\det H \neq 0$ (FULL RANK)

$H_1 = [w(1)] = [0]$ ⇝ WE DIDN'T FIND A CANDIDATE MATRIX SO LET'S GO ON

$$H_2 = \begin{bmatrix} w(1) & w(2) \\ w(2) & w(3) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -1/2 \end{bmatrix} \rightarrow \det H_2 = -1 \neq 0 \checkmark$$

$$H_3 = \begin{bmatrix} w(1) & w(2) & w(3) \\ w(2) & w(3) & w(4) \\ w(3) & w(4) & w(5) \end{bmatrix} = \begin{bmatrix} 0 & 1 & -1/2 \\ 1 & -1/2 & 1/4 \\ -1/2 & 1/4 & -1/8 \end{bmatrix}$$

$$\det H_3 = 0 - 1 \cdot \det \begin{bmatrix} 1 & -1/2 \\ 1/4 & -1/8 \end{bmatrix} - 1/2 \cdot \det \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1/4 \end{bmatrix}$$

$$= -1 \cdot \left[-1/8 + 1/8 \right] - 1/2 \cdot \left[1/4 - 1/4 \right] = 0$$

⇓

$$\boxed{\text{THE SYSTEM ORDER IS } n = 2}$$

STEP # 2 - COMPUTE MATRICES H, G

$$H_{n+1} = \underbrace{\begin{bmatrix} H \\ HF \\ \vdots \\ HF^n \end{bmatrix}}_{3 \times 2} \underbrace{\begin{bmatrix} G & FG & \cdots & F^n G \end{bmatrix}}_{2 \times 3} = \underbrace{O_{n+1}}_{\text{EXTENDED OBSERVABILITY MATRIX}} \cdot \underbrace{R_{n+1}}_{\text{EXTENDED REACHABILITY MATRIX}}$$

LET'S TRY TO FACTORIZE H_{m+1} AS THE PRODUCT OF TWO MATRICES

$$H_{m+1} = \underbrace{\left[\tilde{H}_1 \,|\, \tilde{H}_2\right]}_{3\times 2} \underbrace{\begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \end{bmatrix}}_{2\times 3} = \left[\tilde{H}_1 \,|\, \tilde{H}_2 \,|\, \tilde{H}_3\right]$$

$$= \left[\tilde{H}_1 \,|\, \tilde{H}_2 \,|\, \tilde{H}_3\right] \sim 3\times 3$$

$\tilde{H}_k$: K-TH COLUMN OF H_{m+1} 3×1

THIS HOLDS BECAUSE $\det H_{m+1} = 0$

$$\boxed{a\tilde{H}_1 + b\cdot\tilde{H}_2 = \tilde{H}_3}$$

$\Rightarrow$ LET'S FIND "a" AND "b" IMPOSING

$$a\begin{bmatrix} 0 \\ 1 \\ -1/2 \end{bmatrix} + b\begin{bmatrix} 1 \\ -1/2 \\ 1/4 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/4 \\ -1/8 \end{bmatrix}$$

$\tilde{H}_1$ $\tilde{H}_2$ $\tilde{H}_3$

REMARK: THE COMPUTATION OF "a" AND "b" IS NOT STRICTLY NECESSARY TO RETRIEVE THE VALUES OF H AND G, WHICH DEPEND ONLY ON THE FIRST ROW AND COLUMN OF Q_{m+1} AND R_{m+1}

USING EQUATION # 1 WE GOT: $\boxed{b = -1/2}$

USING EQUATION # 2 WE HAVE: $a + 1/4 = 1/4 \rightarrow \boxed{a = 0}$

USE EQUATION # 3 TO CHECK! $-1/2\cdot 0 - 1/2\cdot 1/4 = -1/8$ ✓

EVENTUALLY WE GOT:

$$H_3 = \begin{bmatrix} \boxed{0 \quad 1} \\ 1 & -1/2 \\ -1/2 & 1/4 \end{bmatrix} \begin{bmatrix} \boxed{1} & 0 & 0 \\ \boxed{0} & 1 & -1/2 \end{bmatrix}$$

REMARK: SINCE $\det H_{m+1} = 0$ THE 3 EQUATIONS ARE LINEARLY DEPENDENT. SO IT IS ENOUGH TO USE ONLY 2 OF THEM, TO COMPUTE "a" AND "b".

THE UNUSED EQUATION STILL HOLDS, SO CAN BE USED TO VERIFY IF THE VALUES OF "a" AND "b" HAVE BEEN CORRECTLY COMPUTED

SO WE CAN COMPUTE H AND G

$$\boxed{H = [0 \quad 1]} \qquad \boxed{G = \begin{bmatrix} 1 \\ 0 \end{bmatrix}}$$

Step # 3 - Compute State matrix F

$$H_{m+1} = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^m \end{bmatrix} \begin{bmatrix} G & FG & \cdots & F^m G \end{bmatrix}$$

m x m

m x m

m x m

3 x 2

2 x 3

- $O(1:end-1, :) \cdot F = O(2:end, :) \Rightarrow F = O(1:end-1,:)^{-1} \cdot O(2:end,:)$

  rows   columns

- $F \cdot R(:, 1:end-1) = R(:, 2:end) \Rightarrow F = R(:,2:end) \cdot R(:,1:end-1)^{-1}$

Using the first formula:

Remark: using the first formula is convenient so that "a" and "b" could not be computed. Using the other formula must give the same result!

$$O = \begin{bmatrix} 0 & 1 \\ 1 & -1/2 \\ -1/2 & 1/4 \end{bmatrix}$$

$$O(1:2,:)^{-1} = \frac{1}{-1} \begin{bmatrix} -1/2 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1/2 & 1 \\ 1 & 0 \end{bmatrix}$$

$$F = \begin{bmatrix} 1/2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1/4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & -1/2 \end{bmatrix} = F$$

Remark: There are indeed ∞ S.S. representations, but the transfer function must be the same!

$$W(z) = H(zI - F)^{-1} G$$

$$= \frac{1}{z(z+1/2)} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} z+1/2 & 0 \\ 1 & z \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{z(z+1/2)} \cdot \begin{bmatrix} 1 & z \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \boxed{\frac{1}{z^2 + 1/2 z}}$$

Same result as before

An alternative Hankel matrix factorization

The Hankel matrix factorization is not unique, consistently with the fact that are ∞ equivalent state space representations of a dynamic system.

In the following, an alternative factorization is proposed. It is based on the "interpretation" of H_{m+1} as a stack of linearly dependent rows:

$\bar{H}$: row of H_{m+1}

$$H_{m+1} = \underbrace{\begin{bmatrix} \quad & \quad \\ \quad & \quad \\ \quad & \quad \end{bmatrix}}_{3\times 2} \underbrace{\begin{bmatrix} \bar{H}_1 \\ \hline \bar{H}_3 \end{bmatrix}}_{2\times 3} = \underbrace{\begin{bmatrix} \bar{H}_1 \\ \hline \bar{H}_2 \\ \hline \bar{H}_3 \end{bmatrix}}_{3\times 3}$$

2 rows of H_{m+1} are arbitrarily selected

Since $\det H_{m+1} = 0$, then the rows are linearly dependent!

$$\boxed{\alpha \bar{H}_1 + \gamma \bar{H}_3 = \bar{H}_2}$$

To impose that $O_{m+1} \cdot R_{m+1} = H_{m+1}$, it must be that

$$= \begin{bmatrix} 1 & 0 \\ \alpha & \gamma \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \bar{H}_1 \\ \hline \bar{H}_3 \end{bmatrix}$$

In order to find "α" and "γ" is imposed:

$$\alpha \cdot \begin{bmatrix} 0 & 1 & -1/2 \end{bmatrix} + \gamma \begin{bmatrix} -1/2 & 1/4 & -1/8 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 & 1/4 \end{bmatrix}$$

Using equation #1 we have: $-\frac{1}{2}\gamma = 1 \Rightarrow \boxed{\gamma = -2}$

Using equation #3 we have: $-\frac{1}{2}\alpha + \frac{1}{4} = \frac{1}{4} \Rightarrow \boxed{\alpha = 0}$

Using equation #2 to check: $\alpha + \frac{1}{4}\gamma = -\frac{1}{2}$

$$0 - \frac{1}{2} \overset{!}{=} -\frac{1}{2} \quad ✓$$

So:

$$H_{m+1} = \begin{bmatrix} 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1/2 \\ -1/2 & 1/4 & -1/8 \end{bmatrix}$$

THE STATE SPACE MATRIX CAN BE COMPUTED:

$$\begin{bmatrix} 0 & 1 & -1/2 \\ 1 & -1/2 & 1/4 \\ -1/2 & 1/4 & -1/8 \end{bmatrix}$$

- $H = \begin{bmatrix} 1 & 0 \end{bmatrix}$
- $G = \begin{bmatrix} 0 \\ -1/2 \end{bmatrix}$

- $$F = \begin{bmatrix} 1 & -1/2 \\ 1/4 & -1/8 \end{bmatrix} \left(\frac{1}{1/2} \begin{bmatrix} 1/4 & -1 \\ 1/2 & 0 \end{bmatrix} \right)^{-1}$$

$$= 2 \cdot \begin{bmatrix} 0 & -1 \\ 0 & -1/4 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 0 & -1/2 \end{bmatrix}$$

THE FOUND STATE SPACE MATRIX IS DIFFERENT FROM THE PREVIOUS ONE. IT IS DUE TO THE DIFFERENT FACTORIZATION OF THE HANKEL MATRIX

HOWEVER THE TRANSFER FUNCTION MUST BE THE SAME. LET'S CHECK:

$$W(z) = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{z(z+1/2)} \begin{bmatrix} 0 & -2 \\ 0 & -1/2 \end{bmatrix} \begin{bmatrix} 0 \\ -1/2 \end{bmatrix}$$

$$= \frac{1}{z^2 + 1/2\, z} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1/4 \end{bmatrix} = \frac{1}{z^2 + 1/2\, z} \quad \checkmark$$

Exercise

Consider the following system:

$$x(t+1) = F\,x(t) + G\,u(t)$$
$$y(t) = H\,x(t)$$

Where:

$$F = \begin{bmatrix} 1 & 9/2 \\ -1 & -7/2 \end{bmatrix} \qquad G = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad H = \begin{bmatrix} 2 & 3 \end{bmatrix} \qquad D = [0]$$

⇒ Compute the system T.F.

$$W(z) = H\,(zI - F)^{-1} \cdot G$$

- $$(zI - F) = \begin{bmatrix} z-1 & -9/2 \\ 1 & z+7/2 \end{bmatrix}$$

- $$(zI - F)^{-1} = \frac{1}{\det(\,)} \begin{bmatrix} z+7/2 & 9/2 \\ -1 & z-1 \end{bmatrix} = \frac{1}{z^2 + 5/2\,z + 1} \begin{bmatrix} & \end{bmatrix}$$

So we have:

$$W(z) = \frac{1}{z^2 + 5/2\,z + 1} \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} z+7/2 & 9/2 \\ -1 & z-1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{z^2 + 5/2\,z + 1} \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 1-z \\ z \end{bmatrix} = \boxed{\frac{z+2}{z^2 + 5/2\,z + 1} = W(z)}$$

⇒ Discuss the stability of the system and also its observability and reachability

System poles: $z^2 + 5/2\,z + 1 = 0$

$$z_{1,2} = -\frac{5}{4} \pm \sqrt{\frac{25}{16} - 1} = -\frac{5}{4} \pm \sqrt{\frac{9}{16}} = \begin{cases} z_1 = -1/2 \\ z_2 = -2 \end{cases}$$

THE SYSTEM IS NOT ASYMPTOTICALLY STABLE, SINCE ONE OF THE POLES IS OUTSIDE THE UNIT CIRCLE.

HOWEVER, IF WE LOOK I/O RELATIONSHIP CLOSER:

$$W(z) = \frac{(z \cancel{+2})}{(z+1/2)(z\cancel{+2})} = \frac{1}{z+1/2} \Rightarrow$$ THE SYSTEM IS EXTERNALLY ASYMPTOTICALLY STABLE

$\Downarrow$

SINCE THERE IS AN HIDDEN PART $\Rightarrow$ THE SYSTEM IS NOT COMPLETELY OBSERVABLE / REACHABLE

LET'S COMPUTE THE OBSERVABILITY AND REACHABILITY MATRICES:

$$\mathcal{O} = \begin{bmatrix} H \\ HF \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ [2 \ \ 3]\begin{bmatrix} 1 & 9/2 \\ -1 & -7/2 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & -3/2 \end{bmatrix}$$

$\det \mathcal{O} = -3+3 = 0 \Rightarrow$ THE SYSTEM IS NOT COMPLETELY OBSERVABLE

$$R = \begin{bmatrix} G & F \cdot G \end{bmatrix} = \begin{bmatrix} -1 & \begin{bmatrix} 1 & 9/2 \\ -1 & -7/2 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} \\ 1 & \end{bmatrix} = \begin{bmatrix} -1 & 7/2 \\ 1 & -5/2 \end{bmatrix}$$

$\det R = 5/2 - 7/2 = -1 \neq 0 \Rightarrow$ THE SYSTEM IS COMPLETELY REACHABLE